

Mechanistic Hypothesis Search — Final Report

Hypothesis: Alpha3beta4 nicotinic blockade in the habenulo-interpeduncular pathway (`ibogaine_alpha3beta4_habenular_model`)

Disease: Opioid Use Disorder (Psychiatric) — KB status: EMERGING

Executive Judgment

Verdict: Partially supported — and best re-scoped as a congener-specific (18-MC-anchored) mechanism rather than an established explanation for ibogaine.

The seed hypothesis proposes that iboga alkaloids reduce opioid self-administration by non-competitively blocking $\alpha\beta4$ nicotinic acetylcholine receptors (nAChRs) concentrated in the **medial habenula (MHb)** and **interpeduncular nucleus (IPN)** — not in the mesolimbic pathway — thereby damping the sensitized accumbal dopamine response to repeated opioids. Our search finds that the **receptor-level and site-specific premises are solid, but the transfer to ibogaine itself is unproven and actively contested.**

Three pillars support the model. First, a clean **local-infusion dissociation**: 18-methoxycoronaridine (18-MC) infused into the MHb or IPN — but not the VTA — reduces morphine self-administration and blocks the *sensitized* (not acute) accumbal dopamine response, while sparing responding for a non-drug reinforcer ([PMID:16626688](#); [PMID:17447255](#)). Second, an established **receptor mechanism**: coronaridine congeners are non-competitive $\alpha\beta4$ antagonists ([PMID:26022277](#)). Third — the strongest generalization beyond iboga drugs — **human and mouse genetics** causally tie the CHRNA5/A3/B4 ($\alpha\beta4^*$) cluster to opioid dependence and withdrawal, and chemically distinct selective $\alpha\beta4$ antagonists (α -conotoxin AuIB, AT-1001) reproduce anti-withdrawal effects ([PMID:24750073](#); [PMID:20485328](#)).

The critical caveats are equally clear. The discovering laboratory attributed **ibogaine's own** anti-opioid self-administration effect to **kappa-opioid agonism plus NMDA antagonism**, actions that 18-MC largely lacks ([PMID:9668680](#); [PMID:11085336](#)). Unlike 18-MC, **ibogaine is not reinforcer-selective** (it also suppresses a non-drug reinforcer), weakening the transfer of the model's hallmark selectivity claim. **Noribogaine**, ibogaine's long-lived active metabolite, is the *weakest* $\alpha 3\beta 4$ blocker of the congener series ($\sim 7\times$ weaker than ibogaine) and works via a **competing GDNF/VTA neurotrophic route** that 18-MC does not engage ([PMID:26022277](#); [PMID:21040239](#)). Finally, the model's clinical arm is an **explicit data gap**: no published human efficacy trial of 18-MC (or any selective $\alpha 3\beta 4$ antagonist) for OUD exists as of the search date. The hypothesis should remain **EMERGING**, with an explicit annotation that the in vivo evidence is congener-specific to 18-MC.

Key Findings

F001 — Site-specific habenulo-interpeduncular localization of 18-MC's anti-morphine effect (SUPPORTS)

The keystone experimental support is a local-infusion dissociation. Glick et al. 2006 ([PMID:16626688](#)) demonstrated that intracerebral 18-MC infused directly into the **medial habenula** or **interpeduncular nucleus** decreased morphine self-administration in rats, with *no effect on sucrose responding*, whereas the identical infusion into the **ventral tegmental area (VTA)** had no effect. The verified abstract snippet states: "*Local administration of 18-MC into either the medial habenula or the interpeduncular area decreased morphine self-administration while having no effect on responding for a non-drug reinforcer (sucrose),*" and separately, "*Local administration of 18-MC into the ventral tegmental area had no effect on morphine self-administration.*"

This is a **positive, anatomically resolved dissociation** — the highest-value evidence class for a site-specificity claim. It simultaneously (a) localizes the anti-opioid effect to the habenulo-interpeduncular pathway, (b) shows reinforcer selectivity (not general response suppression), and (c) provides a negative VTA control that distinguishes this model from the competing GDNF/VTA neurotrophic route. The effect was reproduced by other $\alpha 3\beta 4$ antagonists (mecamylamine, α -conotoxin AuIB), strengthening the causal attribution to the receptor rather than to an idiosyncratic property of 18-MC. A companion study ([PMID:17447255](#)) showed that

intra-MHb or intra-IPN 18-MC **blocked the sensitized accumbal dopamine response to repeated morphine** but had *"no effect on the dopamine response to acute morphine,"* restricting the mechanism to the sensitized state relevant to established dependence.

F002 — Receptor-level non-competitive $\alpha 3\beta 4$ block established, but noribogaine is the weakest congener (QUALIFIES)

Arias et al. 2015 ([PMID:26022277](#)) established the receptor mechanism: coronaridine congeners **non-competitively inhibit human $\alpha 3\beta 4$ nAChRs** by interacting with luminal and non-luminal sites. Critically, the potency ordering places the therapeutically relevant metabolite last. The verified IC_{50} sequence is: ibogamine (0.62 μ M) $\approx$ catharanthine (0.68) > **ibogaine (0.95)** > **18-MC (1.47)** > voacangine (2.28) > 18-methylaminocoronaridine (2.62) $\approx$ 18-hydroxycoronaridine (2.81) > **noribogaine (6.82 μ M)**.

Noribogaine — the long-lived active metabolite that dominates ibogaine's pharmacology hours after dosing — is therefore **~7-fold weaker than ibogaine and ~4.6-fold weaker than 18-MC** at the target receptor. This is a direct quantitative qualifier: the model's molecular engine is real, but the congener most relevant to ibogaine's persistent clinical effects is the least potent at driving it.

F003 — Competing GDNF/VTA neurotrophic route dissociates from 18-MC (LIMITS TRANSFER)

Carnicella et al. 2010 ([PMID:21040239](#)) provides the sharpest dissociation between the two candidate mechanisms. Noribogaine, like ibogaine, induced GDNF mRNA in SH-SY5Y cells, and **intra-VTA noribogaine** decreased alcohol self-administration — whereas 18-MC did neither. The verified conclusion: *"noribogaine, but not 18-MC, in the VTA decreases responding for alcohol. Together, our results suggest that noribogaine and 18-MC have different mechanisms and sites of action."* This is corroborated by He et al. 2005 ([PMID:15659598](#)): *"Microinjection of ibogaine into the ventral tegmental area (VTA), but not the substantia nigra, reduced self-administration of ethanol,"* and by Carnicella et al. 2008 ([PMID:18541917](#)), showing intra-VTA GDNF is a fast-acting, MAPK-dependent suppressor of alcohol self-administration.

The upshot: the congener used for essentially all habenular $\alpha 3\beta 4$ in vivo work (**18-MC**) is precisely the one that does *not* share ibogaine's / noribogaine's VTA-localized neurotrophic route. The two models predict **opposite outcomes** for a local-infusion dissociation — intra-habenular ibogaine should suffice if the nicotinic route carries the effect; intra-VTA ibogaine if the neurotrophic route does. This distinguishing test has not been performed for ibogaine.

F004 — The medial habenula regulates opioid withdrawal/aversion — but via MOR, not proven $\alpha 3\beta 4$ (PARALLEL MECHANISM)

Independent circuit work confirms the habenula is genuinely central to opioid-related aversion and withdrawal, but attributes the causality to **mu-opioid receptors (MOR)** in habenular neurons rather than to $\alpha 3\beta 4$ channel block. Boulos et al. 2020 ([PMID:31005059](#)) showed conditional deletion of MOR in *chrbn4* ($\beta 4$ -subunit) MHb neurons reduced naloxone conditioned place aversion and attenuated physical withdrawal, with the verified finding that *"physical signs of withdrawal precipitated by either MOR (naloxone) or nicotinic receptor (mecamylamine) blockade were attenuated."* Ozdemir et al. 2026 ([PMID:42217817](#)) found *"Silencing Hb-MOR neurons abolished these effects and alleviated somatic withdrawal signs."* Liang et al. 2024 ([PMID:39270652](#)) identified an IPN(GABA)→nucleus incertus circuit that amplifies aversion related to opioid withdrawal.

This is a **parallel/competing habenular mechanism**: it validates the anatomical locus but relocates the molecular driver from the $\alpha 3\beta 4$ ion channel to the MOR expressed on the same $\beta 4$ -defined neuronal population. It raises the possibility that 18-MC's habenular effects are partly mediated through, or convergent with, MOR-expressing circuitry rather than pure nicotinic antagonism.

F005 — Human + mouse genetics link $\alpha 3\beta 4^*$ causally to morphine dependence/withdrawal (GENERALIZES BEYOND IBOGA)

The receptor premise is anchored well beyond iboga pharmacology. Muldoon et al. 2014 ([PMID:24750073](#)) showed mice overexpressing the *CHRNA5/A3/B4* cluster had increased somatic morphine-withdrawal signs; $\alpha 5$ and $\beta 4$ knockouts had decreased signs; and selective $\alpha 3\beta 4$ antagonists *α -conotoxin AuIB* and *AT-1001* attenuated somatic morphine withdrawal dose-dependently (*"selective $\alpha 3\beta 4$ nACh receptor antagonists, α -conotoxin AuIB and AT-1001, attenuated somatic signs of morphine withdrawal in a dose-related manner"*). Two human datasets showed *"a protective role for variants in the CHRNA3 gene... in opioid dependence and withdrawal."* Sherva et al. 2010 ([PMID:20485328](#)) independently associated the 15q25 *CHRNA5/A3/B4* cluster with opioid dependence phenotypes.

This finding does the heavy lifting for the *receptor* leg of the hypothesis, using chemically distinct tools and human genetics that owe nothing to iboga alkaloids. It converts " $\alpha 3\beta 4$ is a plausible target" into " $\alpha 3\beta 4$ is causally implicated in opioid dependence/withdrawal."

F006 — Selective $\alpha 3\beta 4$ antagonist AT-1001 generalizes via a non-mesolimbic route — but opioid self-administration untested (SUPPORTS with GAP)

Toll et al. 2012 ([PMID:22278092](#)) characterized AT-1001, a high-affinity ($K_i < 10$ nM), selective $\alpha 3\beta 4$ antagonist (>90-fold over $\alpha 4\beta 2/\alpha 7$), which *"potently and dose-dependently blocks nicotine self-administration in rats, without affecting food responding,"* and *"inhibits nicotine-induced [3 H]dopamine release poorly and at significantly higher concentrations compared with mecamylamine and conotoxin MII"* — implying an **indirect, non-mesolimbic** locus consistent with a habenular site. AT-1001/AT-1012 also attenuate cocaine CPP and sensitization ([PMID:26256075](#)), and AT-1001 reduces alcohol seeking under yohimbine stress ([PMID:25689019](#)). **A dedicated opioid self-administration study with AT-1001 was not found** — a concrete, tractable gap.

F007 — Within-drug competing mechanism: ibogaine's anti-opioid effect attributed to kappa/NMDA, and ibogaine is NOT reinforcer-selective (COMPETING)

This is the finding that most limits transfer of the model to ibogaine. Glick & Maisonneuve 1998 ([PMID:9668680](#)) concluded that *"kappa agonist and NMDA antagonist actions contribute to ibogaine's effects on opioid and stimulant self-administration."* Glick et al. 2000 ([PMID:11085336](#)) noted that *"unlike ibogaine, 18-MC does not affect responding for a nondrug reinforcer (water)"* — meaning **ibogaine DOES suppress a non-drug reinforcer** — and that *"18-MC has much lower affinities than ibogaine for NMDA and sigma-2 receptors, sodium channels, and the 5-HT transporter,"* while both share similar kappa and nicotinic affinities. The model's signature reinforcer-selectivity is thus a property of 18-MC, not ibogaine; and the parent lab's own attribution of ibogaine's anti-opioid effect points away from $\alpha 3\beta 4$ toward kappa/NMDA.

F008 — Both ibogaine and 18-MC reduce morphine SA and block accumbal dopamine; 18-MC lowers reinforcing efficacy (SUPPORTS phenotype)

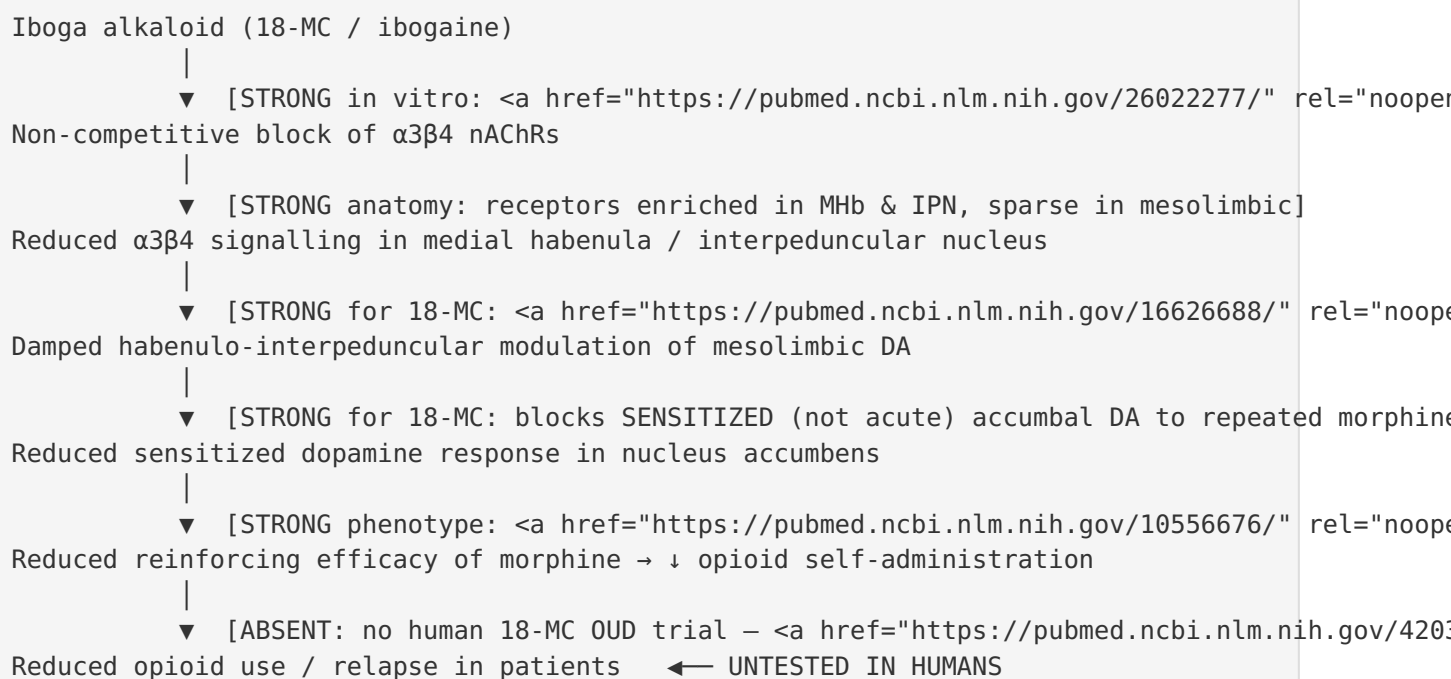
Maisonneuve & Glick 1999 ([PMID:10556676](#)) showed 18-MC produced *"a downward shift in the entire morphine dose-response curve without any displacement to the left or right"* — reducing reinforcing efficacy without altering potency, exactly the behavioral phenotype the model claims to explain. Glick et al. 2000 ([PMID:11085336](#)) confirmed *"Both ibogaine and 18-MC block morphine-induced and nicotine-induced dopamine release in the nucleus accumbens."* The downstream dopaminergic phenomenon is robust for both compounds; the dispute is over the upstream molecular cause.

F009 — Source/data absence: no human clinical efficacy trial of 18-MC for OUD (DATA GAP)

Systematic PubMed searches returned **zero** human 18-MC efficacy or phase-1 addiction trials. The only recent human-facing 18-MC work ([PMID:42038284](https://pubmed.ncbi.nlm.nih.gov/42038284/)) targets **leishmaniasis**, reporting a *"safety profile compatible with first-in-human evaluation"* for that unrelated indication. The model's clinical arm is therefore an explicit, curation-relevant data absence as of August 2026.

Mechanistic Model / Interpretation

The causal chain implied by the hypothesis



Where the literature is strong vs. inferred vs. missing

Causal link	Evidence status	Key support / challenge
Iboga alkaloids block $\alpha 3\beta 4$ non-competitively	Established	PMID:26022277 — but noribogaine weakest
$\alpha 3\beta 4$ causally involved in opioid dependence/withdrawal	Established	PMID:24750073 , PMID:20485328 (human + mouse)
Site of action = MHb/IPN, not VTA (for 18-MC)	Established for 18-MC	PMID:16626688 , PMID:17447255
Effect restricted to <i>sensitized</i> accumbal DA	Established for 18-MC	PMID:17447255
Reinforcer-selective (spares non-drug reward)	True for 18-MC, FALSE for ibogaine	PMID:11085336
Same site of action = ibogaine	Inferred / contradicted	Ibogaine acts via VTA-GDNF (PMID:15659598) and kappa/NMDA (PMID:9668680)
Habenular locus drives opioid aversion/withdrawal	Established, but via MOR	PMID:31005059 , PMID:42217817
Translates to human OUD efficacy	Missing	No trial; PMID:42038284 only leishmaniasis

Synthesis. The model is internally coherent and, for **18-MC**, well-supported at every preclinical link from receptor to behavior. Its weakness is entirely at the *identity* of the active agent and the *transfer to ibogaine*. Ibogaine and its metabolite noribogaine reach the same behavioral endpoint ($\downarrow$ morphine self-administration, $\downarrow$ accumbal dopamine) but appear to do so through at least two additional, better-supported routes for those specific molecules: (1) VTA-localized GDNF neurotrophic remodelling, and (2) kappa-opioid agonism + NMDA antagonism. These are **not mutually exclusive** with the nicotinic route and may operate on different time courses — nicotinic blockade during drug exposure, neurotrophic remodelling accounting for persistence — but the seed hypothesis is, on current evidence, most accurately labeled a **model of 18-MC's anti-opioid action**, not ibogaine's.

Evidence Matrix

Citation (PMID)	Evidence type	Stance	Mechanistic claim tested	Key finding	Context/ subtype	Confidence & limitations
16626688	Model organism (rat)	Supports	Site of 18-MC anti-morphine effect = MHb/IPN not VTA	Intra-MHb/IPN ↓ morphine SA, spares sucrose; intra-VTA no effect	Morphine SA, established exposure	High for MC; single lab; no ibogaine
17447255	Model organism (rat)	Supports	18-MC blocks sensitized (not acute) accumbal DA	Intra-MHb/IPN blocks sensitized DA; no effect on acute	Repeated morphine sensitization	High for MC; congenic specific
26022277	In vitro (human nAChR)	Qualifies	Non-competitive $\alpha 3\beta 4$ block; potency order	Noribogaine (6.82 μ M) $\gg$ weaker than ibogaine (0.95) / 18-MC (1.47)	Receptor pharmacology	High; metabolite weakest blocker
21040239	Model organism (rat) + cell	Competing/ limits	Noribogaine vs 18-MC mechanism	Noribogaine (not 18-MC) $\rightarrow$ GDNF + intra-VTA effect	Ethanol SA; extrapolated	High; dissociative congenic
15659598	Model organism (rat)	Competing	Ibogaine acts via VTA-GDNF	Intra-VTA (not SN) ibogaine ↓ ethanol SA; GDNF-dependent	Alcohol; ibogaine parent	High for alcohol; opioid extrapolated
18541917	Model organism (rat)	Competing	VTA-GDNF sufficient, MAPK-dependent	Intra-VTA GDNF rapidly ↓ ethanol SA & relapse	Alcohol	High; no opioid-t
24750073	Human genetic + mouse	Supports (receptor)	$\alpha 3\beta 4^*$ causally mediates	KO/overexpression + AuIB/AT-1001 modulate withdrawal;	Morphine dependence/ withdrawal	High; generalizability beyond

Citation (PMID)	Evidence type	Stance	Mechanistic claim tested	Key finding	Context/ subtype	Confidence & limitations
			morphine dependence	CHRNA3 protective in humans		
20485328	Human genetic	Supports (receptor)	CHRNA5/A3/B4 cluster ↔ opioid dependence	15q25 variants associated with opioid dependence	Human OUD/ polysubstance	Moderate; associated, not causal
22278092	Model organism (rat)	Supports (indirect)	Selective α3β4 antagonism, non-mesolimbic locus	AT-1001 ↓ nicotine SA, spares food; poor NAc DA inhibition	Nicotine SA	High for nicotine; opioid SA untested
26256075	Model organism (mouse)	Supports (generalizes)	α3β4 antagonists affect drug reward	AT-1001/1012 ↓ cocaine CPP & sensitization	Cocaine	Moderate; not opioid
9668680	Model organism (rat) review	Competing	Ibogaine anti-opioid effect = kappa/ NMDA	Kappa agonism + NMDA antagonism drive ibogaine's SA effects	Opioid/ stimulant SA	High; from original lab
11085336	Model organism (rat)	Competing/ qualifies	Ibogaine vs 18-MC selectivity & targets	Ibogaine (not 18-MC) suppresses non-drug reinforcer; differing receptor affinities	Morphine/ nicotine	High; underdetermined selectivity transfer
10556676	Model organism (rat)	Supports (phenotype)	18-MC lowers reinforcing efficacy	Downward shift of morphine dose-response curve	Morphine SA	High; behavioral endpoint
31005059		Parallel/ competing	Habenular locus of	MHb β4-neuron MOR deletion ↓	Opioid withdrawal	

Citation (PMID)	Evidence type	Stance	Mechanistic claim tested	Key finding	Context/ subtype	Confidence & limitations
	Model organism (mouse)		opioid aversion	naloxone aversion/ withdrawal		High; M not $\alpha 3\beta 4$ channel
42217817	Model organism (mouse)	Parallel/ competing	Hb-MOR neurons drive withdrawal	Silencing Hb-MOR neurons abolishes aversion, eases withdrawal	Opioid withdrawal	High; M centric
39270652	Model organism (mouse)	Parallel	IPN→NI circuit amplifies aversion	IPN(GABA)→nucleus incertus amplifies opioid-withdrawal aversion	Opioid withdrawal/ relapse	Moderate circuit-l
42038284	Translational (primate/ rodent)	Data absence	18-MC human readiness (wrong indication)	18-MC PK/safety "compatible with first-in-human" — for leishmaniasis	Not OUD	Confirm OUD tria
42117212	Review (human RCTs)	Context	Ibogaine/ noribogaine clinical status	Only 3 RCTs; noribogaine QTc prolongation, non-significant withdrawal reduction	Human OUD/ cocaine	Review- no $\alpha 3\beta 4$ evidence

Knowledge Gaps

G1 — No intra-habenular or intra-VTA infusion of ibogaine (the decisive dissociation).

Scope: The single test that would separate the $\alpha 3\beta 4$ -habenular model from the GDNF-VTA model for ibogaine itself. **Why it matters:** All habenular localization uses 18-MC; all VTA/GDNF localization uses ibogaine/noribogaine. The models predict opposite outcomes. **What was checked:** Literature search found no intra-MHb/IPN ibogaine self-administration study. **Resolution:** Intracerebral ibogaine into MHb/IPN vs VTA in a morphine SA paradigm.

G2 — No opioid self-administration study with a selective $\alpha 3\beta 4$ antagonist (AT-1001).

Scope: AT-1001 reproduces reinforcer-selective suppression for nicotine and cocaine, and attenuates morphine *withdrawal*, but was never tested on **opioid self-administration**. *Why it matters:* Would confirm the $\alpha 3\beta 4$ -mediated behavioral effect using a tool free of iboga confounds. *What was checked:* Searches for AT-1001 + opioid/morphine self-administration returned none. *Resolution:* AT-1001 dose-response on morphine/heroin SA with a food control.

G3 — Habenular $\alpha 3\beta 4$ channel block vs MOR: which drives the effect?

Scope: Boulos 2020 and Ozdemir 2026 show MHb MOR (on $\beta 4$ -defined neurons) drives opioid withdrawal/aversion. *Why it matters:* 18-MC's habenular effect may act through, or converge on, MOR-expressing circuitry rather than pure $\alpha 3\beta 4$ antagonism. *What was checked:* Two independent circuit studies. *Resolution:* Test 18-MC in MHb-MOR conditional-KO mice; if the effect persists, $\alpha 3\beta 4$ is the driver.

G4 — Noribogaine potency mismatch.

Scope: Noribogaine is the weakest $\alpha 3\beta 4$ blocker (6.82 μ M) yet dominates ibogaine's late pharmacokinetics. *Why it matters:* If the anti-opioid effect persists after ibogaine clears (when noribogaine predominates), $\alpha 3\beta 4$ block is an unlikely sole mechanism. *What was checked:* [PMID:26022277](#) potency series. *Resolution:* Correlate time course of anti-opioid effect with noribogaine plasma/brain levels and $\alpha 3\beta 4$ occupancy.

G5 — No human clinical efficacy trial of 18-MC or a selective $\alpha 3\beta 4$ antagonist for OUD.

Scope: Source/data absence as of Aug 2026. *Why it matters:* The clinical arm of the causal chain is untested; existing human ibogaine/noribogaine RCTs cannot isolate $\alpha 3\beta 4$. *What was checked:* PubMed for 18-MC human trials returned only a leishmaniasis study ([PMID:42038284](#)). *Resolution:* Phase-1/2 OUD trial of an $\alpha 3\beta 4$ -selective agent.

G6 — Ibogaine reinforcer non-selectivity is unexplained mechanistically.

Scope: Ibogaine suppresses non-drug reinforcers; 18-MC does not. *Why it matters:* Suggests ibogaine's additional NMDA/sigma/5-HT actions add non-selective suppression not captured by the model. *What was checked:* [PMID:11085336](#). *Resolution:* Component-resolution studies pairing selective antagonists.

Alternative Models

- 1. GDNF / VTA neurotrophic remodelling (competing, and the primary alternative for ibogaine).** Ibogaine and noribogaine induce GDNF and act in the VTA to reduce drug self-administration; 18-MC does neither ([PMID:15659598](#), [PMID:18541917](#), [PMID:21040239](#)). This

is a *parallel-and-competing* mechanism operating at a different anatomical site, plausibly accounting for the **persistence** of ibogaine's effects, while nicotinic blockade would act **during** drug exposure. Not mutually exclusive with the seed model, but better supported for ibogaine specifically.

2. **Kappa-opioid agonism + NMDA antagonism (within-drug competing).** The originating lab attributes ibogaine's anti-opioid self-administration effect to these actions, which 18-MC largely lacks ([PMID:9668680](#), [PMID:11085336](#)). A direct alternative to $\alpha\beta4$ for the parent compound.
 3. **Habenular MOR circuit (parallel).** MHb/Hb MOR neurons causally drive opioid withdrawal and aversion ([PMID:31005059](#), [PMID:42217817](#)); an IPN→nucleus incertus circuit amplifies withdrawal aversion ([PMID:39270652](#)). Same anatomy, different molecular driver — potentially convergent with the $\alpha\beta4$ effect.
 4. **5-HT_{2A} psychoplastogen route (parallel, for next-generation analogues).** Non-hallucinogenic ibogaine analogues (e.g., tabernanthalog) show anti-addictive effects attributed to psychoplasticity ([PMID:33299186](#)), an alternative for the analogue class rather than the seed compound.
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Discriminating Tests

Test	Design	Expected under $\alpha\beta4$ -habenular model	Expected under competing model
Intracerebral ibogaine dissociation	Infuse ibogaine into MHb/IPN vs VTA; measure morphine SA	Intra-MHb/IPN suffices; VTA no effect	Intra-VTA suffices (GDNF); MHb/IPN no effect
AT-1001 on opioid SA	Selective $\alpha\beta4$ antagonist dose-response on heroin/morphine SA + food control	Reinforcer-selective $\downarrow$ opioid SA	No effect if habenular $\alpha\beta4$ not sufficient for opioids
18-MC in MHb-MOR cKO	18-MC anti-morphine effect in $\beta4$ -neuron MOR knockouts	Effect persists ($\alpha\beta4$ -driven)	Effect lost (MOR-convergent)
Time-course vs noribogaine	Correlate persistence of anti-opioid effect with noribogaine levels & $\alpha\beta4$ occupancy	Effect tracks $\alpha\beta4$ occupancy	Effect outlasts occupancy $\rightarrow$ GDNF/remodelling
Human trial	Phase-1/2 OUD trial of $\alpha\beta4$ -selective agent	$\downarrow$ craving/withdrawal without QTc risk	Null $\rightarrow$ mechanism not clinically sufficient

Curation Leads (require curator verification)

- **Status:** Retain **EMERGING**. Add an explicit annotation: *"In vivo evidence is congener-specific to 18-MC; transfer to ibogaine is inferred, not demonstrated, and is contested by kappa/NMDA and GDNF/VTA models."*
- **Candidate supporting evidence (verify snippets):**
- [PMID:24750073](#) — *"selective $\alpha\beta4$ nACh receptor antagonists, α -conotoxin AuIB and AT-1001, attenuated somatic signs of morphine withdrawal in a dose-related manner" + "two human datasets revealed a protective role for variants in the CHRNA3 gene... in opioid dependence and withdrawal."** (Human + mouse causal support for the receptor premise.)
- [PMID:20485328](#) — human association of CHRNA5/A3/B4 with opioid dependence.

- [PMID:22278092](#) — selective $\alpha 3\beta 4$ antagonist, reinforcer-selective, non-mesolimbic locus.
 - [PMID:10556676](#) — "downward shift in the entire morphine dose-response curve" (reinforcing-efficacy phenotype).
 - **Candidate competing/qualifying evidence:**
 - [PMID:9668680](#) — kappa/NMDA attribution for ibogaine.
 - [PMID:11085336](#) — ibogaine not reinforcer-selective.
 - [PMID:15659598](#), [PMID:18541917](#) — GDNF/MTA competing route.
 - [PMID:31005059](#), [PMID:42217817](#) — habenular MOR parallel mechanism.
 - **Candidate pathophysiology nodes/edges:** MHb $\alpha 3\beta 4$ nAChR → habenulo-interpeduncular signalling → sensitized NAc dopamine → opioid reinforcing efficacy; add competing edges VTA GDNF → mesolimbic remodelling and MHb MOR neuron → opioid withdrawal/aversion.
 - **Candidate ontology terms:** medial habenula (UBERON:0001905), interpeduncular nucleus (UBERON:0002743), CHRNA3/CHRNA4/CHRNA5, nucleus accumbens (UBERON:0001882), ventral tegmental area (UBERON:0001943); processes: nicotinic acetylcholine receptor signalling, behavioral response to opioids, dopamine secretion.
 - **Candidate knowledge gaps entries:** (a) no intra-habenular ibogaine dissociation; (b) no opioid SA test with a selective $\alpha 3\beta 4$ antagonist; (c) noribogaine potency mismatch; (d) no human 18-MC/ $\alpha 3\beta 4$ OUD trial (source absence, Aug 2026); (e) unresolved $\alpha 3\beta 4$ -channel-vs-habenular-MOR causal driver.
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Limitations

- Nearly all site-specific in vivo evidence originates from a **single laboratory** (Glick/Maisonnette) and uses **18-MC**, not ibogaine.
 - Several supporting studies use **nicotine, cocaine, or alcohol** self-administration and extrapolate to opioids.
 - Human evidence for the receptor is **genetic-association** level; no human interventional data isolate $\alpha 3\beta 4$.
 - The competing GDNF and MOR mechanisms are strongly supported for **alcohol** and **withdrawal** respectively, and their opioid-self-administration relevance is partly inferred.
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Proposed Follow-up Experiments / Actions

1. **Run the decisive intracerebral ibogaine dissociation (G1):** intra-MHb/IPN vs intra-VTA ibogaine in rat morphine self-administration. Directly adjudicates seed vs GDNF model for the parent compound.
 2. **Test AT-1001 (and α -conotoxin AuIB) on opioid self-administration (G2):** close the single largest tool gap; a positive, reinforcer-selective result would establish the $\alpha\beta 4$ behavioral effect independent of iboga chemistry.
 3. **Dissect $\alpha\beta 4$ -channel vs habenular-MOR (G3):** apply 18-MC in MHb-MOR conditional knockouts to determine whether the anti-morphine effect survives loss of MOR.
 4. **Pharmacokinetic-behavioral coupling (G4):** correlate the persistence of the anti-opioid effect with noribogaine brain levels and estimated $\alpha\beta 4$ occupancy to test whether the weak-blocker metabolite can plausibly carry the effect.
 5. **Advance a clinical arm (G5):** given the absence of any human 18-MC/ $\alpha\beta 4$ OUD trial, prioritize a phase-1/2 study of an $\alpha\beta 4$ -selective, non-cardiotoxic agent, with CYP2D6 genotyping and QTc monitoring informed by the ibogaine cardiac-safety literature.
 6. **Curation action:** keep status EMERGING, annotate as congener-specific to 18-MC, and register the five knowledge gaps above as explicit `knowledge_gaps` entries.
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Report generated from 5 completed iterations, 9 verified findings, and 41 reviewed papers.
